

QIAN CHEN

AI4Medical & Healthcare · Federated Learning · Pre-Trained Model Fusion/Merging · Trustworthy AI · Foundation Model

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EDUCATION

Institute of Computing Technology, Chinese Academy of Sciences (ICT, CAS)

Beijing, China

Doctor of Philosophy - Computer Science. Supervisor: Prof. Yiqiang Chen

2020.09 - 2025.06

Lanzhou Jiaotong University

Lanzhou, China

Master of Engineering - Computer Science. Supervisor: Prof. Xuejun Zhang

2017.09 - 2020.06

Cooperation: Worked with Prof. Yiqiang Chen as a visiting student at ICT, CAS from Mar. 2017 to Sep. 2020.

Tianjin Agricultural University

Tianjin, China

Bachelor of Engineering - Computer Science

2012.09 - 2016.06

RESEARCH INTERESTS

- **Trustworthy Federated learning:** Pre-trained models fusion, transfer learning, privacy, fairness.
- **Multi-source Data Fusion:** Pre-trained models fusion, multi-modal fusion, sensor network.
- **Foundation Model:** Foundation model understanding & fusion, model federated network, agentic workflows.
- **Application:** Medical image analysis, wearable healthcare, emotion and human activity recognition.

SELECTED PUBLICATIONS

- [1] [TKDE'24] PrivFusion: Privacy-Preserving Model Fusion via Decentralized Federated Graph Matching
Qian Chen, Yiqiang Chen, Xinlong Jiang, et al.
IEEE Transactions on Knowledge and Data Engineering (CCF A / Top, JCR Q1, IF: 10.4)
- [2] [ICDE'24] Model Trip: Enhancing Privacy and Fairness in Model Fusion across Multi-Federations for Trustworthy Global Healthcare
Qian Chen, Yiqiang Chen, Bingjie Yan, Xinlong Jiang, ..., Lixin Fan, Qiang Yang
IEEE International Conference on Data Engineering (CCF A / Top-tier, Oral)
- [3] [CIKM'24] Buffalo: Biomedical Vision-Language Understanding with Cross-Modal Prototype and Federated Foundation Model Collaboration
Bingjie Yan, Qian Chen, Yiqiang Chen, Xinlong Jiang, et al.
ACM International Conference on Information and Knowledge Management (CCF B / CORE A, Oral)
- [4] [TKDE'25] Ten Challenging Problems in Federated Foundation Models
Tao Fan, Hanlin Gu, Xuemei Cao, Chee Seng Chan, Qian Chen, Yiqiang Chen, ..., Lixin Fan, Qiang Yang.
IEEE Transactions on Knowledge and Data Engineering (CCF A / Top, JCR Q1, IF: 10.4)
- [5] [JCST'25] MoFedNet: Semantic Link is All Model Need
Yiqiang Chen, Xinlong Jiang, Qian Chen, et al.
Journal of Computer Science and Technology (Accepted; Best Paper Award at CCF Computility 2025; First author as student, main contributor, CCF B)
- [6] [AAAI'25] Mitigating Pervasive Modality Absence through Multimodal Generalization and Refinement
Wuliang Huang, Yiqiang Chen, Xinlong Jiang, Chenlong Gao, Teng Zhang, Qian Chen, et al.

- [7] [ICME'25] FairFHTL: Achieving Task-Agnostic Fairness in Federated Hetero-Task Learning
Teng Zhang, Yiqiang Chen, Xinlong Jiang, Wuliang Huang, Qian Chen, et al.
IEEE International Conference on Multimedia & Expo (CCF B)
- [8] [MM'24] Correlation-Driven Multi-Modality Graph Decomposition for Cross-Subject Emotion Recognition
Wuliang Huang, Yiqiang Chen, Xinlong Jiang, Chenlong Gao, Qian Chen, et al.
ACM International Conference on Multimedia (CCF A / Top-tier)
- [9] [Cell Patterns'24] FedEYE: A Scalable and Flexible End-to-end Federated Learning Platform for Ophthalmology
Bingjie Yan, Danmin Cao, Xinlong Jiang, Yiqiang Chen, ..., Qian Chen, et al.
Patterns (Cell Press, JCR Q1, IF: 7.4)
- [10] [TIST'24] Survey on knowledge distillation for large language models: methods, evaluation, and application

Chuanpeng Yang, Yao Zhu, Wang Lu, Yidong Wang, Qian Chen, et al.
ACM Transactions on Intelligent Systems and Technology (JCR Q1, IF: 6.6)
- [11] [JCST'24] Fedbone: Towards large-scale federated multi-task learning
Yiqiang Chen, Teng Zhang, Xinlong Jiang, Qian Chen, et al.
Journal of Computer Science and Technology (CCF B)
- [12] [TOMM'23] GJFusion: A Channel-Level Correlation Construction Method for Multimodal Physiological Signal Fusion
Wuliang Huang, Yiqiang Chen, Xinlong Jiang, Teng Zhang, Qian Chen.
ACM Transactions on Multimedia Computing, Communications and Applications (JCR Q1, IF: 6.0)
- [13] [BIBM'24] EyeGraphGPT: Knowledge Graph Enhanced Multimodal Large Language Model for Ophthalmic Report Generation
Zhirui Wang, Xinlong Jiang, Chenlong Gao, Weiwei Dai, Bingyu Wang, Bingjie Yan, Qian Chen, et al.
IEEE International Conference on Bioinformatics and Biomedicine (CCF B, Oral)
- [14] [ICPADS'23] FedTAM: Decentralized Federated Learning with a Feature Attention Based Multi-teacher Knowledge Distillation for Healthcare
Tingting Mou, Xinlong Jiang, Jie Li, Bingjie Yan, Qian Chen, et al.
IEEE International Conference on Parallel and Distributed Systems (CCF C, Oral)
- [15] [NCA'22] A differentially private indoor localization scheme with fusion of WiFi and bluetooth fingerprints in edge computing
Xuejun Zhang, Fucun He, Qian Chen, et al.
Neural Computing and Applications (JCR Q2)
- [16] [UIC'19] Differential Privacy-based Indoor Localization Privacy Protection in Edge Computing
Xuejun Zhang, Qian Chen, Xiaohui Peng, Xinlong Jiang.
IEEE International Conference on Ubiquitous Intelligence and Computing (CCF C, Oral)

- [17] [TKDE'25] AdaEns: Dual-Loop Adaptive Alignment, Selection and Weighting Ensemble for noisy domain generalization
Wang Lu, Qian Chen, Teng Zhang, et al.
IEEE Transactions on Knowledge and Data Engineering (Co-First Author, Under Review)
- [18] [TKDE'25] Across Data Heterogeneity: Dynamic Neighbors Graph for Decentralized Personalized Federated Learning
Tingting Mou, Xinlong Jiang, Qian Chen, et al.
IEEE Transactions on Knowledge and Data Engineering (Under Review)
- [19] [TIST'25] FHBench: Towards Efficient and Personalized Federated Learning for Multimodal Healthcare
Penghao Wang, Zhaoyang Ma, Qian Chen, et al.
ACM Transactions on Intelligent Systems and Technology (Under Review)
- [20] [Annals of Medicine'25] Collaborative Artificial Intelligence for the Diagnosis and Management of Acute Ischemic Stroke
Zhiqiang Fan, Qian Chen, Wang Lu, et al.
Annals of Medicine (Co-First Author, Under Review)
- [21] [Manuscript] MetaFusion: Redefining Model Weights as Distinct Modalities for Interpretability-Guided Federated Model Fusion
Qian Chen, Yiqiang Chen, et al.
(Manuscript in preparation)
- [22] [Manuscript] RobustFusion: Federated Model Fusion with Unbalanced Optimal Transport for Robustness against Data and Noise Heterogeneity
Qian Chen, Yiqiang Chen, et al.
(Manuscript in preparation)
- [23] [Manuscript] FusionHive: A Scheduling Methods for Federated Model Network Services
Qian Chen, Yiqiang Chen, et al.
(Manuscript in preparation)
- [24] [Manuscript] Model Journey: Path-Aware Federated Model Fusion via Hypergraph Optimization
Qian Chen, Yiqiang Chen, et al.
(Manuscript in preparation, Extend by ICDE'24)
- [25] [Patent] A Federated Model Network-Based Multi-Center Medical Task Model Construction Method
Yiqiang Chen, Qian Chen, Xinlong Jiang, Chenlong Gao, Teng Zhang, Wuliang Huang, Bingjie Yan.
CN Patent 202411354767.4 (Pending, future research focus)
- [26] [Patent] A Cross-Modal Transfer Learning Method
Xinlong Jiang, Qian Chen, Yiqiang Chen, Teng Zhang, Yang Gu.
CN Patent 113139664B (Granted)
- [27] [Patent] A Federated Learning Method and System for Sample Sparsity
Yiqiang Chen, Qian Chen, Xinlong Jiang, Xin Qin.
CN Patent 113128701A

- [28] **[Patent] A Method and System for Data Analysis Based on Federated Learning**
Xinlong Jiang, Qian Chen, Yiqiang Chen, Teng Zhang, Yongbin Wang, Zhongping Zhang, Yishan Xiao.
CN Patent 113052333A
- [29] **[Patent] A Method, Device, Equipment, and Medium for Updating Fundus Image Recognition Models**
Yiqiang Chen, Xinlong Jiang, Qian Chen, Weiwei Dai.
CN Patent 117197115A
- [30] **[Patent] A Dynamic Location Privacy Protection Method and Device in Edge Computing Environments**
Wei Zeng, Xinlong Jiang, Qian Chen, Zhiwen Pan, Hui Zhang.
CN Patent 110611667B (Granted)
- [31] **[Patent] A Cross-Domain Adaptive Graph Convolutional Balanced Transfer Learning Method and System**
Yiqiang Chen, Chenlong Gao, Xinlong Jiang, Qian Chen.
CN Patent 113128667B (Granted)
- [32] **[Patent] A Federated Construction Method for Medical Task Models and Medical Image Classification Methods**
Yiqiang Chen, Teng Zhang, Xinlong Jiang, Chenlong Gao, Qian Chen, Wuliang Huang, Zhen Yan.
CN Patent 117612709A
- [33] **[Patent] A Differential Private Multi-Source Wireless Signal Fingerprint Fusion Indoor Localization Method in Edge Computing Architecture**
Xuejun Zhang, Qian Chen, Junda Bao, Fucun He, Jiyang Gai, Xiaogang Du, Haiyan Huang, Tao Ju.
CN Patent 111988845B (Granted)
- [34] **[Patent] A Federated Learning-Based Indoor Localization Privacy Protection Method for Edge Computing**
Xuejun Zhang, Fucun He, Qian Chen, Jiyang Gai, Junda Bao, Tao Ju, Haiyan Huang, Xiaogang Du.
CN Patent 111866869B (Granted)

SELECTED AWARDS

2025 Outstanding Graduate (Top 5%), awarded by the Beijing Municipal Education Commission, China	Beijing
2024 National Scholarship (Top 2%), awarded by the Ministry of Education, P. R. China	Beijing
2024 Merit Student (Top 15%), awarded by the University of Chinese Academy of Sciences	Beijing
2023 Merit Student (Top 15%), awarded by the University of Chinese Academy of Sciences	Beijing
2022 The First Prize Scholarship (Top 10%), awarded by the University of Chinese Academy of Sciences	Beijing
2020 Outstanding Thesis Award (Top 3%), awarded by the Lanzhou Jiaotong University	Lanzhou
2020 Outstanding Graduate Award (Top 7%), awarded by the Lanzhou Jiaotong University	Lanzhou
2018 National Second Prize (Top 2%), the National Undergraduate Internet of Things (IoT) Design Contest	Wuxi

PROJECTS

Research on Incremental Model Fusion Methods for Multi-Center Medical Collaboration General Program

National Natural Science Foundation of China 2025.01 - 2028.12

- I focus on developing advanced fusion techniques to streamline and secure data sharing across medical centers. This initiative targets enhancing collaborative efficiency and ensuring data privacy in fields like medical imaging and genomics. By integrating intelligent model selection and multitasking strategies, the project aims to optimize the utilization of diverse model resources, fostering robust, secure, and fair cross-center collaborations in a globalized medical environment.*

Research on Federated Transfer Learning Methods for Early Warning of Parkinson's Disease General Program

National Natural Science Foundation of China 2020.01-2023.12

- I developed federated learning approaches that integrated multi-source data on Parkinson's without requiring data sharing, enhancing personalized prediction accuracy through adaptive transfer learning. This work effectively addressed the challenges of early warning models, improving prediction precision for individual cases of Parkinson's disease.*

Development of an Intelligent Big Data Platform for Eye Health Management Key Regional STS

Chinese Academy of Sciences 2021.01-2022.12

- I engaged in comprehensive data processing, analysis, and modeling tasks. My responsibilities included designing algorithms for efficient data handling, implementing analytical frameworks to interpret complex ocular health datasets, and constructing predictive models to enhance early diagnosis and treatment strategies. This work underpinned the system's ability to autonomously monitor and manage eye health, leveraging big data to provide actionable insights and significantly improve patient outcomes.*

SERVICES

The China-Japan-Korea (CJK) Forum on Aging and Disability CNCC

Assistant Organizer 2019.10 - Present

- Contributed to the CJK Forum, an international platform under the China National Computer Congress (CNCC) dedicated to advancing AI and technology for elderly care and disability support. The forum has engaged over 5,000 participants, including representatives from renowned institutions such as the University of Tokyo, KAIST, NEC, Samsung Electronics and others.*

The IEEE Standard Development IEEE Standard

Participant 2022.04 - Present

- Contributed to the development of IEEE Guide for an Architectural Framework for Explainable Artificial Intelligence (P2894), IEEE Guide for an Architectural Framework and Application for Collaborative Edge Computing (P2961) and IEEE Guide for K-12 Educational Knowledge Graphs (P2807.6), which aim to establish international standards in the field of intelligent systems and computing technologies.*

Review Contributions

As an active peer reviewer, I've maintained high standards of scientific rigor, contributing significantly to the editorial processes of prominent IEEE journals such as IEEE Transactions on Neural Networks and Learning Systems (TNNLS) and IEEE Transactions on Knowledge and Data Engineering (TKDE). My role also involves reviewing submissions for esteemed conferences like the International Conference on Multimedia and Expo (ICME), Ubiquitous Intelligence and Computing (UIC), AAAI Conference on Artificial Intelligence (AAAI), and the International Joint Conference on Artificial Intelligence (IJCAI). In addition, I serve as a Program Committee member for both AAAI-26 and the AAAI 2026 AI Alignment Track. This experience has sharpened my critical evaluation skills, kept me updated on cutting-edge research, and enhanced my contributions to the advancement of the field.